

Test Plan — Swag Labs (<https://www.saucedemo.com>)

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Role applied for: QA (manual)

Version: 1.0

Application under test: Swag Labs demo e-commerce site

1. Objective

Verify that the core shopping journey (login → browse → cart → checkout → logout) works correctly for each provided user role, and find defects that affect users, data accuracy, or trust in the product.

2. Scope

In scope

Area	Features covered
Authentication	Login, logout, locked-out user, invalid credentials, session handling, direct-URL access without login
Product browsing	Inventory list, images, names, prices, descriptions, sorting (A–Z, Z–A, price low–high, high–low), product detail page
Cart	Add, remove, cart badge count, cart persistence, continue shopping, empty cart
Checkout	Step 1 (customer info), Step 2 (overview, totals, tax), completion page, cancel at each step
Navigation	Burger menu (All Items, About, Logout, Reset App State), footer social links
Non-functional (basic)	Response time per user, responsive layout, browser compatibility

Out of scope

- Real payment processing (none exists on the site)
- Load/stress testing, penetration testing (only basic observations of client-side security)
- Backend/API testing (no access)
- Automation (pure manual QA track)

3. Types of Testing

Type	What I will do
Functional	Verify each feature works as specified, per user role
UI / Visual	Check alignment, images vs. product names, icons, fonts, colors, consistency
Negative	Wrong/empty credentials, empty checkout fields, invalid characters, locked-out user
Edge cases / Boundary	Empty cart checkout, all 6 items in cart, very long text in fields, whitespace-only input, special characters, emoji, leading/trailing spaces
Exploratory	Session-based charters (see §9) — unscripted, guided by curiosity and risk
State / Session	Cart after logout/login, browser Back/Forward/Refresh at each step, multiple tabs, cookie/localStorage behavior
Cross-browser	Same smoke suite on multiple browsers (see §4)
Usability / Accessibility (basic)	Keyboard-only navigation (Tab/Enter), error message clarity, focus order
Performance (basic)	Compare login/page-load time for <code>performance_glitch_user</code> vs <code>standard_user</code>

4. Test Environment

Item	Details
Primary browser	Brave (latest) on Windows 11
Secondary browsers	Microsoft Edge (latest)
Mobile / responsive	Chrome DevTools device emulation: iPhone 12 (390×844), Pixel 7 (412×915), iPad Air (820×1180); also resize desktop window to 1366×768 and 1920×1080
Tools	Browser DevTools (Console, Network, Application → Cookies/Local Storage), screenshot tool, screen recorder (Loom), stopwatch
Network	Home broadband; one pass with DevTools throttling = "Slow 3G" for performance_glitch_user

5. Test Data

Username	Password	Purpose
standard_user	secret_sauce	Baseline / expected behavior
locked_out_user	secret_sauce	Must be blocked with a clear error
problem_user	secret_sauce	Known-faulty UI/behavior
performance_glitch_user	secret_sauce	Slow responses

Additional data I will create

Data	Values
Invalid logins	standard_user/wrong, wrong_user/secret_sauce, empty/empty, username only, password only, STANDARD_USER (case), standard_user (spaces), ' OR 1=1 --
Checkout names	John, A, 100-char string, 12345, !@#\$\$%, (spaces), José Ñandú, x, emoji
Postal codes	1100, ABCDE, 00000, -1, 30-digit string, empty, spaces

6. Entry / Exit Criteria

- **Entry:** Site reachable, all 4 usernames available, browsers installed, test plan reviewed.
- **Exit:** All planned test cases executed at least once; all Critical/Major bugs documented with reproduction steps and evidence; summary of results written.

7. Test Cases

ID	Title	Preconditions	Steps	Expected Result	Priority
TC-01	Login with valid credentials	On login page	1. Enter standard_user 2. Enter secret_sauce 3. Click Login	Redirected to /inventory.html; 6 products shown; cart empty	High
TC-02	Locked-out user sees error	On login page	1. Enter locked_out_user / secret_sauce 2. Click Login	Stays on login page; error "Sorry, this user has been locked out."; no session created	High
TC-03	Login with invalid password / empty fields	On login page	1. Enter valid username + wrong password → Login 2. Clear all fields → Login 3. Password only → Login	Appropriate error for each ("Username and password do not match...", "Username is required", "Password is required"); fields highlighted; error dismissible	High

ID	Title	Preconditions	Steps	Expected Result	Priority
TC-04	Direct URL access without login	Logged out	Open <code>/inventory.html</code> , <code>/cart.html</code> , <code>/checkout-step-one.html</code> directly	Redirected to login with error "You can only access ... when you are logged in"	High
TC-05	Sort products	Logged in as <code>standard_user</code>	Select each of the 4 sort options	List order matches option (A-Z, Z-A, price asc, price desc); images/prices stay attached to correct names	Medium
TC-06	Add / remove item from inventory page	Logged in, empty cart	1. Click Add to cart on each of 6 items 2. Verify badge = 6 3. Click Remove on each	Button toggles Add↔Remove; badge increments/decrements correctly; badge disappears at 0	High
TC-07	Product image/name/price consistency	Logged in	Compare each inventory tile with its product detail page (click name and image)	Same name, image, price, description on both pages	High
TC-08	Cart contents and persistence	Logged in	1. Add 2 items 2. Open cart 3. Continue Shopping 4. Return to cart 5. Refresh	Items, quantity, prices unchanged each time	Medium
TC-09	Complete checkout (happy path)	1+ item in cart	Cart → Checkout → enter First/Last/Postal → Continue → verify overview → Finish	Overview shows right items; Item total + 8% tax = Total; "Thank you for your order!" page; cart badge cleared	Critical
TC-10	Checkout field validation	1+ item in cart, on step one	Submit with: all empty; only first name; only last name; no postal; special characters/numbers/whitespace in names	Each missing field → specific error. Invalid/whitespace-only input should be rejected	High
TC-11	Checkout with empty cart	Cart empty	Go to <code>/cart.html</code> → click Checkout	Checkout should be blocked or warn the user	High
TC-12	Totals calculation	Add items with varied prices	Complete to overview step; calculate manually	Item total = sum of prices; tax = 8% rounded to 2 dp; total = item total + tax	High
TC-13	Logout and session handling	Logged in, cart has items	Burger → Logout; press browser Back; log in again	Back doesn't expose protected pages; cart behavior after re-login is consistent and expected	High
TC-14	Reset App State	Items added	Burger → Reset App State	Cart cleared, badge removed, all buttons return to "Add to cart" without needing a refresh	Medium
TC-15	performance_glitch_user timing	Fresh session	Log in, add item, go to cart, checkout; time each action vs <code>standard_user</code>	Functional behavior identical; delays noted; UI gives feedback during wait	Medium
TC-16	problem_user full journey	Logged in as <code>problem_user</code>	Repeat TC-05, 06, 07, 09	Behavior identical to <code>standard_user</code> (any difference = defect)	High
TC-17	Responsive layout	Any user	View inventory, cart, checkout at 390px / 820px / 1366px widths	No overlap, no horizontal scroll, buttons tappable	Medium
TC-18	Keyboard accessibility	Any user	Complete login + add to cart + checkout using only Tab / Enter / Space	Logical focus order, visible focus, all actions reachable	Low

8. Risk Assessment

Risk area	Likelihood of defects	Impact	Reason / how I'll mitigate
Checkout form (validation)	High	High	Revenue-critical; known fields with no/weak validation. Boundary + invalid-input tests
<code>problem_user</code> behavior (cart, images, sort, inputs)	High	High	Account is built to be faulty; compare against <code>standard_user</code> line by line
Cart state & session (persistence, reset, logout)	High	Medium	State is kept client-side; test across logout/login, refresh, multiple tabs
Empty-cart / no-item checkout	High	High	Business logic gap; test directly
Price/tax calculation	Medium	High	Rounding errors cause financial/legal issues
Sorting	Medium	Medium	Easy to break; verify data stays paired with correct item
<code>performance_glitch_user</code>	High	Medium	Intentional delay; check feedback to user and double-submit risk
Auth / access control	Medium	High	Client-side session cookie; test direct URLs, cookie edits
Cross-browser / responsive layout	Medium	Medium	Minor CSS differences usually appear in Firefox/mobile widths
Invalid item IDs / URL tampering	Medium	Medium	Product page uses query-string IDs

9. Exploratory Testing Charters (time-boxed, 20–30 min each)

1. **Explore** cart state **with** logout/login and multiple users **to discover** whether cart data leaks between accounts.
2. **Explore** the checkout forms **with** odd inputs (spaces, symbols, emoji, very long text) **to discover** validation gaps and layout breaks.
3. **Explore** the product detail page **with** modified URLs (`?id=999`, `?id=-1`, `?id=abc`) **to discover** error handling.
4. **Explore** burger menu and footer links **with** each user **to discover** broken links/navigation.
5. **Explore** the session cookie/local storage in DevTools **to discover** access-control weaknesses.
6. **Explore** browser Back/Forward/Refresh at every step **to discover** state or duplicate-order issues.

10. Deliverables & Reporting

- Bug reports (see `02_Bug_Report.md`) with ID, severity, priority, steps, expected/actual, environment, evidence.
- Severity scale: **Critical** (blocks core flow / data loss / security), **Major** (feature broken, workaround exists), **Minor** (cosmetic or low impact).

11. Assumptions & Limitations

- Demo site; no real backend or payments.
- Requirements are inferred from standard e-commerce behavior since no spec was provided.
- Tested manually only; no automation is part of this submission.